

Junseok Lee

ROBOTICS · AUTONOMOUS DRIVING

☎ (+82) 010-8794-1762 | ✉ ryan082688@gmail.com | 🏠 jundduck.github.io | 📷 jundduck | 🌐 jundduck

Personal Profile

My research interest lies in developing autonomous driving and mobility systems that meaningfully improve people's lives. While improving technical performance is very important, I focus more how these technologies can be applied in real world environments and what practical benefits can be provided.

Having experienced Paraplegia in the past, I deeply understand how mobility limitations can affect one's independence and quality of life. This experience motivates me to develop a stronger interest in mobility technologies that can improve safety and accessibility not only for the general public, but also for transportation vulnerable individuals, including people with mobility impairments.

Education

Yonsei University

B.S. in Civil & Environmental Engineering

Seoul

Mar. 2022 - Aug. 2026

Research Experience

MOTIF DRIVE

E2E Autonomous Driving Perception Team

Seoul, Republic of Korea

July. 2025 - Aug. 2025

Computational Intelligence Lab (CILAB)

Undergraduate Research Internship, Research on AI-based control systems and embedded robotics using deep learning (Advisor: Eun-Tai Kim)

Seoul, Republic of Korea

Feb. 2025 - Jun. 2025

High-Dimensional Accurate Measurement and Intelligent Sensing Lab (DAMIS)

Worked on vision-based sensing using high-resolution cameras and 3D Reconstruction (Advisor: Jae-Sang Hyun)

Seoul, Republic of Korea

Jul. 2024 - Dec. 2024

Honors & Awards

Oct 2025 **Excellence Award**, AI Convergence Intensive Major Competition

Yonsei University

Oct 2025 **Encouragement Award**, HL FMA Autonomous Driving Competition

HL Mando, Klemove

Sep 2025 **Grand Prize**, Korea Robot Aircraft Competition

Sacheon, KRAC

Jun 2025 **Excellence Award**, QRC Korea (Quadruped Robot Challenge)

Jeonju, ICROS 2025

Nov 2024 **Grand Prize**, Yonsei-Asan UnivCT Climate Tech Start-Up

Yonsei University

Publications

Uncertainty Triggered Multi-Rate Re-ranking to Counter Uncertainty-Induced Conservatism in High-Speed World-Model-Based Trajectory Selection

IEEE Intelligent Vehicles Symposium

(IV) 2026 Workshop

First Author (JunSeok Lee*, Jaein Jang*, [Siho Kim])

- Introduced an uncertainty-aware re-evaluation strategy that selectively re-ranks top candidate trajectories only under high uncertainty, improving trajectory decisiveness while maintaining computational efficiency.

A Fusion-Based Curb Detection Framework Using LiDAR and Driving Trajectory for Path Planning in CARLA

KRoC 2026 Conference

First Author (JunSeok Lee*, Siho Kim)

- Recovering missing curb annotations in CARLA by fusing 3D LiDAR point clouds with driving trajectories, using weak-perspective projection and trajectory-guided tiling/post-processing to scale to large maps and improve path planning.

Semantic Segmentation-Based Autonomous Flying Drone’s Victim Recognition
Algorithm in the Complex Terrain

ICROS 2025 Conference

First Author (Lee Jun-Seok*, Jeon Yu-jin, Kim Eun-tai)

- Proposed a Semantic Segmentation-based algorithm to identify victims in cluttered outdoor environments using hand-designed Autonomous Drone.

Extracurricular Projects

Hyundai Motor Scholarship Program

Yonsei University

Member

Oct. 2024 – Aug.2025

- Participating in various autonomous driving and robot competitions sponsored by Hyundai Motor’s contract department (Mobility System Engineering Lab with Hyundai Motors)

XYZ Innovation

Seoul

Robotics Developer (Supported by Hyundai Asan Foundation)

Jun. 2024 – Jan. 2025

- Climate Tech Start-up, Providing income in the form of credits to the owners through the forest carbon offset system

Yonsei Drone

Yonsei University

Member

Sep 2024 – Sep 2025

- Developed Autonomous Flight Drone systems with expertise in flight control, End-to-End navigation for real-world environments.

Skills

Programming Language	C++, Python
DevOps	Git, Docker, ROS2, ROS
Languages	Korean, English